

Olivier Belan

PhD Student @ UKC & Research Assistant @ ICNS

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TECHNICAL SKILLS

Languages: Python, Cython, C/C++

Libraries/Frameworks: Cuda, Jax, Pytorch, Numba, Ray

Developer Tools: Git, Docker

OPEN PROJECTS

Evo-Sim

github.com/OlivierBelan/Evo-Sim

- Simulator for Spiking & Artificial Neural Networks supporting GA, NEAT, HyperNetwork, MAP-Elites, NSLC, CMA-ES, NES, OpenES, ARS, ES-HyperNEAT.
- **Tech:** Python, Cython, C/C++, CUDA, Ray, Numba, NumPy, PyTorch

SNN-Sim

github.com/OlivierBelan/SNN-Sim

- Clock-based Spiking Neural Network simulator with Cython and CUDA acceleration.
- **Tech:** Python, Cython, C/C++, CUDA, NumPy

SSM-Sim

github.com/OlivierBelan/SSM-Sim

- State-Space Model Reinforcement Learning simulator (S4, S5, S4-Event-Based) implemented in JAX with PPO. (*Work in progress*)
- **Tech:** Python, JAX

EXPERIENCE

Research Assistant

Mar. 2025 - Mar. 2026

International Centre for Neuromorphic Systems (ICNS)

Sydney, Australia

- Working under Dr Saeed Afshar, Dr Yeshwanth Ravi Theja Bethi, Dr Alexandre Marcireau and Dr Sergio Chevtchenko. Research in Spiking Neural Networks, NeuroEvolution, agent control and neuromorphic systems.
- Developing SNN architectures tailored for low-latency, real-time edge deployment.

Research Assistant & Software Developer

Jan. 2025 - Feb. 2026

University of Kent

Canterbury, United Kingdom

- ML researcher and software developer on the *Personalising Domiciliary Dementia Care* project, supervised by Dr Daniel Soria.
- Designing personalised care-matching logic and data-driven app modules.

Researcher & Graduate Teaching Assistant

Sep. 2021 - Present

University of Kent

Canterbury, United Kingdom

- Member of the Artificial Intelligence and Data Analytics (AIDA) research group.
- Teaching Assistant for undergraduate and postgraduate modules: Cognitive Neural Networks, Machine Learning, Deep Learning, Programming for AI, Advanced OOP, Functional Programming.

Research and Development – Software Developer

Jun. 2020 - Feb. 2022

Pasteur Institute & Brain & Spine Institute

Paris, France

- Contributor to the *Toi-même* ecosystem for mental disorder research.
- Enhanced mobile and backend tooling for large-scale clinical and behavioural data collection.

Teaching Assistant

Sep. 2019 - Mar. 2020

Web@cademie by EPITECH

Paris, France

- Teaching programming and project management; supporting first- and second-year students in developing technical and professional skills.

Teaching Assistant

Sep. 2019 - Mar. 2020

Coding Academy by EPITECH

Paris, France

- Similar teaching responsibilities as Web@cademie, adapted to a different student profile and academic structure.

Software Developer

Jul. 2018 - Dec. 2018

DT Consulting

Paris, France

- Contributed to the roadmap and development of a C# ticketing solution for sports venues, improving usability and feature coverage.

PAPERS

First-Author Papers

Event-Based State-Space-Model for RL (in preparation)

International Centre for Neuromorphic Systems

Sydney, Australia

Fragmented Fourier Series to optimise Spiking Neural Network (in preparation)

University of Kent

Canterbury, United Kingdom

Neuron Energy Distribution to optimise Spiking Neural Network (in preparation)

University of Kent

Canterbury, United Kingdom

Spikes Augmentation Decoder for Spiking Neural Network (in preparation)

University of Kent

Canterbury, United Kingdom

Combinatorial Encoder for Spiking Neural Network (in preparation)

University of Kent

Canterbury, United Kingdom

Benchmark: NeuroEvolution and Spiking Neural Network in RL (in preparation)

University of Kent

Canterbury, United Kingdom

Co-Authored Papers

Quantization Hardware-Aware State-Space-Model (in preparation, 2nd author)

International Centre for Neuromorphic Systems

Sydney, Australia

Survey: UAV and Spiking Neural Networks (in preparation, 4th author)

International Centre for Neuromorphic Systems

Sydney, Australia

EDUCATION

University of Kent

Canterbury, United Kingdom

Philosophiae Doctor, School of Computing

2021 - 2026

- NeuroEvolution, Spiking Neural Networks, Reinforcement Learning, Agent-Control, guided by Dr. Dominique Chu and Dr. Marek Grzes.

University of Kent

Canterbury, United Kingdom

Master of Science in Advanced Computer Science (Computational Intelligence)

2020 - 2021

- Master thesis: NeuroEvolution and Spiking Neural Network (Distinction), Supervised by Dr. Dominique Chu
- MSc specialized curriculum focusing on Artificial Intelligence (Distinction)

Epitech

Paris, France

School of Computing

2020 - 2021

- Final Project and Dissertation: "Polaris/Toi-même" - Developed an ecosystem for mental disorder research, in collaboration with the Pasteur Institute and Brain and Spine Institute, Paris. Achieved finalist status in a nationwide competition among Epitech schools - (Distinction).

Epitech

Paris, France

Bachelor's Degree, School of Computing

2017 - 2020

- Specialized in low-level programming, with a focus on developing a strong foundational skill set in C and C++ languages.

CONFERENCES

NCN, Neuromorphic Computing Netherlands (poster)

September 2024

Eindhoven University of Technology

IEEE/ACM ICONS, International Conference On Neuromorphic Systems (poster) August 2024

George Mason University

FUNDING & SCHOLARSHIP

PhD Scholarship - Graduate Teaching Assistant - 20 000£

Sep 2021 - Sep 2024

University of Kent

Canterbury, United Kingdom

SERVICE AND VOLUNTARISM

UKC French Society - Head of Logistic

2023 - 2024

University of Kent

Canterbury, United Kingdom

- The French Society at the University of Kent serves as a vibrant hub for those interested in French culture. As Head of Logistics, I planned and executed events, managed resources, liaised with vendors, and ensured memorable experiences for attendees

Coding Club By Epitech - Teaching

2017 - 2020

Epitech

Paris, France

- Coding Club is a recurrent workshop that introduces programming and development. The sessions are led in a fun way through various activities, such as developing video games, solving puzzles or cyber security activities, and more

LANGUAGES & PASSION

Languages: French (native), English (Fluent) (but with a nice French accent, of course)

Passion: Apéro, Wine, Martial Arts, Rugby, Football (Soccer), Guitar

REFERENCES

Yeshwanth Ravi Theja Bethi : y.bethi@westernsydney.edu.au

Saeed Afshar : s.afshar@westernsydney.edu.au

Dominique Chu: d.f.chu@kent.ac.uk

Marek Grzes: m.grzes@kent.ac.uk

Daniel Soria: d.soria@kent.ac.uk

Fabrice Marco: fabrice.marco@epitech.eu